Lab 2 — Requirements Analysis of a Real Project

Design of Software System · Turin Polytechnic University in Tashkent

Goal of this lab

In Lecture 2 we learned what a requirement is, the difference between functional and non-functional requirements, the five techniques of collecting them, and the six properties that every written requirement must have. In this lab you do the work of an analyst on one project from the beginning: you read the words of a customer, you turn them into a numbered list, you find what the customer forgot to say, and you repair the requirements written badly by somebody else.

- No programming in this lab. Everything is analysis and writing.
- Work alone. You may discuss ideas with a classmate, but the text must be your own.
- Submit one file: `Surname_Name_Lab2.pdf`.

Total: 20 points. Expected working time: 2–3 hours.

The customer

Case study: the TTPU student print service.

The university opens a small print room for students. The head of the department explains it like this:

"A student sends us a document and our operator prints it. Every student gets 100 free pages at the beginning of the semester. When the free pages are finished, printing costs 200 soum per page, and a colour page costs twice as much as a black-and-white one. One job may not be longer than 50 pages, otherwise the queue stops for everybody else. Each job gets its own number, and the student can cancel it while it is still waiting; once it is printed, it is too late. Our operator needs to see the jobs that are waiting. At the end of the day I want to know how many pages were printed and how much money we collected. And please make it simple — our operator is 58 years old and has never used anything except Excel."

Task 1 — Write the requirements (5 points)

Read the words of the customer and turn them into a numbered list of requirements. Present them in a table with four columns:

ID	Requirement (use the word "shall")	F / N	How it would be tested
FR-01	The system shall give every new student 100 free pages per semester.	F	Register a new student and check that 100 free pages are shown.
…	…	…	…

- At least **ten functional** requirements taken from the text. Every sentence uses "shall" and contains a number wherever the customer gave one.
- At least **four non-functional** requirements covering **four different qualities** (performance, security, usability, reliability, portability, maintainability). Only one of them is hidden in the text of the customer — find it. The other three you propose yourself as the analyst, and they must be realistic for a university print room.

- The fourth column is one line only. If you cannot describe the test, the requirement is not testable and must be rewritten before you submit it.

Task 2 — Plan the requirement collection (3 points)

The text above is what one person said in five minutes. A real analyst does not stop there.

2.1 For each of the five techniques from the lecture — interview, questionnaire, observation, document analysis, prototyping — write one line: **from whom** you would collect, and **what exactly** you expect to learn in this project that you do not know yet.

2.2 Name the technique you would use **first**, and explain in two sentences why that one and not another.

2.3 Write **eight questions** you would ask the customer in the interview. They must be questions whose answers are not in the text above. Group them under three headings of your own, for example *money*, *rules*, *users*.

Task 3 — Repair the requirements of another analyst (5 points)

A colleague sent you the eight requirements below for the same project. Every one of them breaks at least one of the six properties: clear, testable, complete, consistent, necessary, traceable.

```
a)  The printing must be fast.
b)  The system should handle the students.
c)  A student can print up to 50 pages, and long documents of 80 pages are also allowed.
d)  The operator interface must be modern and beautiful, like the interfaces of 2025.
e)  The system must calculate the price.
f)  Printing shall be free.
g)  The program should probably send some kind of message when something happens.
h)  The system shall use a PostgreSQL database with an index on the job_id column.
```

For each of the eight statements write three things:

3.1 Which property or properties it breaks, and why — one sentence.

3.2 The corrected requirement, with an ID, the word "shall" and a number where a number is possible.

3.3 The one-line test of your corrected version.

Two of these eight statements cannot be repaired by rewriting them, because the problem is not the words. Find them, say what is really wrong, and explain what you would do with them.

Task 4 — What the customer did not say (3 points)

The description of the customer looks complete, but it is not. A good analyst finds the holes before the design starts.

4.1 Write **six situations** that the customer did not describe and that the system will meet in the first week of work. For example: the student sends a job, pays, and never comes to take the printed pages.

4.2 For each situation write the requirement you would add to close it.

4.3 Choose the one you consider the most dangerous for the project, and explain in three sentences what would happen if nobody noticed it until the deployment phase. Use the cost chart of the lecture in your explanation.

Task 5 — The project through the six phases (4 points)

Fill in this table for the print service project. Write real sentences about this project, not the general definitions from the slides.

Phase	What exactly is done in THIS project	What comes out of it	Who does it
1. Requirement gathering			
2. Feasibility study			
3. Design			
4. Coding			
5. Testing			
6. Deployment and maintenance			

5.1 Fill in all four columns for all six phases.

5.2 Under the table answer: the university gives you 6 weeks and no money for a second programmer. Does the feasibility study say yes or no, and on which of the five feasibility types does your answer depend?

Grading

Task	What is checked	Points
1	10 functional and 4 non-functional requirements, IDs, "shall", numbers, a real test in the fourth column, four different qualities	5
2	Five techniques with a source and a purpose, a justified first choice, eight grouped questions that are not answered in the text	3
3	Eight statements: the broken property named, the corrected version, the test; the two unrepairable ones found	5
4	Six missing situations, six added requirements, one reasoned answer with the cost chart	3
5	The table filled for six phases, and the feasibility answer	4
	Total	**20**

Submission. One file, `Surname_Name_Lab2.pdf`, sent on Telegram to @ethz_uz before the announced deadline. Answers copied from another student receive zero for both students. Work sent after the deadline is not accepted.